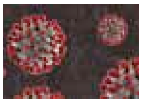


Know Your Enemy
Here's a complete day-by-day breakdown of COVID-19 symptom progression. **https://dgit.in/dec20-21**

Parasitic Infusion
The tiny 'theragrippers' latch onto the intestinal tract, to slowly release drugs into the body. **https://dgit.in/dec20-22**

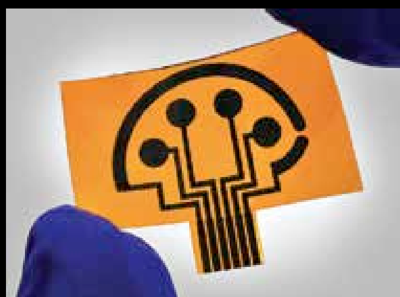
begins to play a role only after populations develop an immunity to the virus.

Researchers from the University of Texas used real world data over a period of five months, from March to July to check if weather had any effect on human beings. The combination of temperature and humidity conditions were reduced to a single number. Additionally, human behaviour was also tracked, using cellphone data to investigate travel habits. The two data sets were independent in the sense that the researchers did not factor in how the weather might affect the behaviour of the people. Both these factors were then measured up next to the spread of the coronavirus in different countries, and multiple factors were ranked in terms of their importance in the outcome. These calculations showed that in all countries, the weather had only a 3 percent role to play, and there was no indication that a particular kind of weather favoured or curtailed the spread of the virus. The most important factors were long distance trips, spending time away from home, population and urban density, with the last contributing 13 percent. Both the simulation and the observations indicate that the weather conditions are not important for the spread of the virus, when humans have not built up immunity.

While most of the research is focused on the biochemical properties of the coronavirus, a group of researchers from MIT used atom level computer simulations to probe the physical characteristics of coronavirus spikes, the structures that give these viruses their characteristic appearance. The study investigated the structures on a number of coronaviruses, including SARS-CoV, MERS-CoV, SARS-CoV-2 and one of its mutations. The research found a positive correlation between the differences in the vibrational characteristics and the infectivity of the virus. The spikes on the proteins can vibrate to fit the ACE2 receptor on cells that allow the virus inside the cell. Once inside, the virus proceeds to hijack the cellular machinery of the cell to begin making copies of itself. The mechanism by which the virus gains access to the cell is similar to how jiggling a key of the same shape in a lock can unlock it. The research shows that it might be possible to deduce the infectivity of novel coronaviruses as soon as they emerge, by just observing their physical characteristics.

NEW TESTING APPROACHES

Caltech professor Wei Gao has come up with a low cost sensor that combines graphene with biological molecules to detect traces of the virus, antibodies created by the immune system as a response to infection, as well as chemical markers of the infection. Digit has previously covered Gao's work in the 2019 December issue, which described a sensor that



Credit: Caltech

The low cost Caltech sensor chip

could pick up indications of cardiovascular, liver or kidney diseases by detecting small concentrations of trace molecules in samples of sweat. This is the only platform that can provide information about the disease in three different types of data. The test takes only a few minutes to perform, and works on small amounts of saliva or blood, without needing the assistance of a medical health professional. At present the technology is in the prototyping stage, with only a small number of devices being tested in a lab, but the indications are that it is a reliable test. The widespread dissemination of this cheap test, or other similar at home testing kit, can be crucial in curtailing the further spread of the virus. This is because there are spreaders who may be infecting others, while being asymptomatic themselves. This test is sensitive enough to detect the infection even when the patient is asymptomatic.

Researchers from Harvard have developed a chip that is about half the size of a credit card, contains a complex network of microfluidic channels, and can detect the presence of the novel coronavirus within half an hour. The test detects trace amounts of RNA, telltale signature of the virus that it leaves behind in biological material. Once a sample such as a nasal swap is collected, the genetic material is processed and amplified in the microfluidic samples. Then, the CRISPR-Cas12 enzyme (a close relative of CRISPR-Cas9) is used to check if any of the genetic material is from the novel coronavirus. If the virus is detected, the enzyme gets activated and triggers fluorescent probes that causes the samples to glow. In its current state, the DNA amplification is performed externally, but the researchers hope to include it within the chip over the next few months. The approach is not specific to Covid-19 and can potentially be used to investigate the presence of other microbes as well.

Researchers from the University of Texas have developed a new antigen test that can detect antibodies in a large volume of samples and very quickly. Conventionally, the best available serological test looks for the number of virus neutralizing antibodies in the blood, which is a time consuming and expensive test, and requires many days for interpretation. The researchers used an alternative testing method, called enzyme-linked immunosorbent assays or ELISA, that looked for antibodies against specific SARS-CoV-2 spike proteins, instead of all SARS-CoV-2 antibodies. The amount of the antibodies for these specific spike proteins within the blood stream is then used as a proxy for the total amount of antibodies in the blood. A test based on a little more than 2,800 blood samples showed that the ELISA based test had at least an 80 percent probability of matching the results of the VN test. d